

The Perceptron

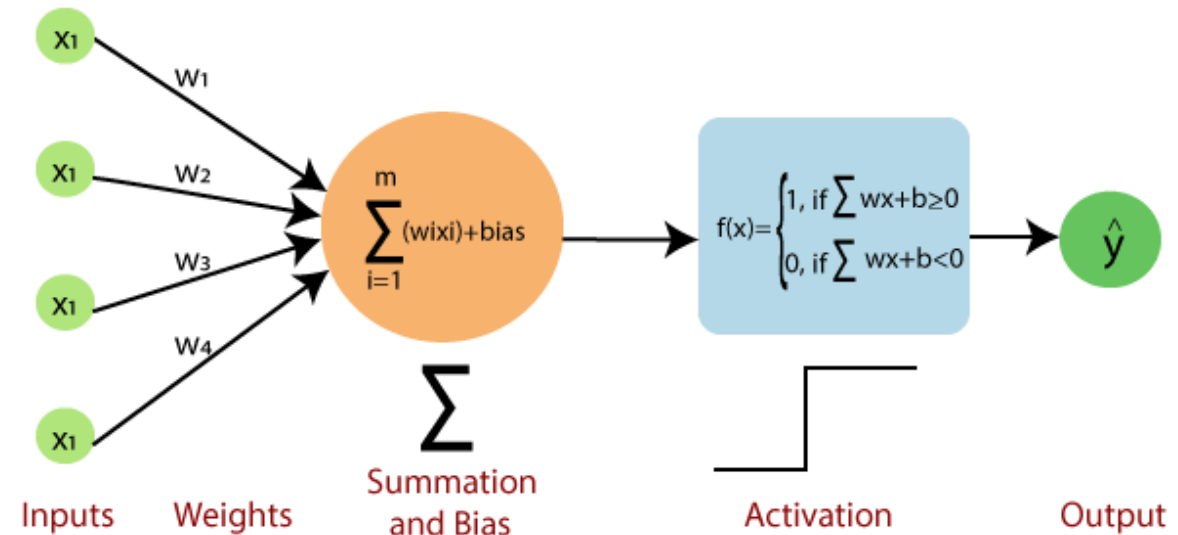
- Input: feature vector $x \in \mathbb{R}^m$
- Label: $y \in \{0, 1\}$
- Model:

$$\hat{y} = \sigma(w^T x + b)$$

- **Update rule:**

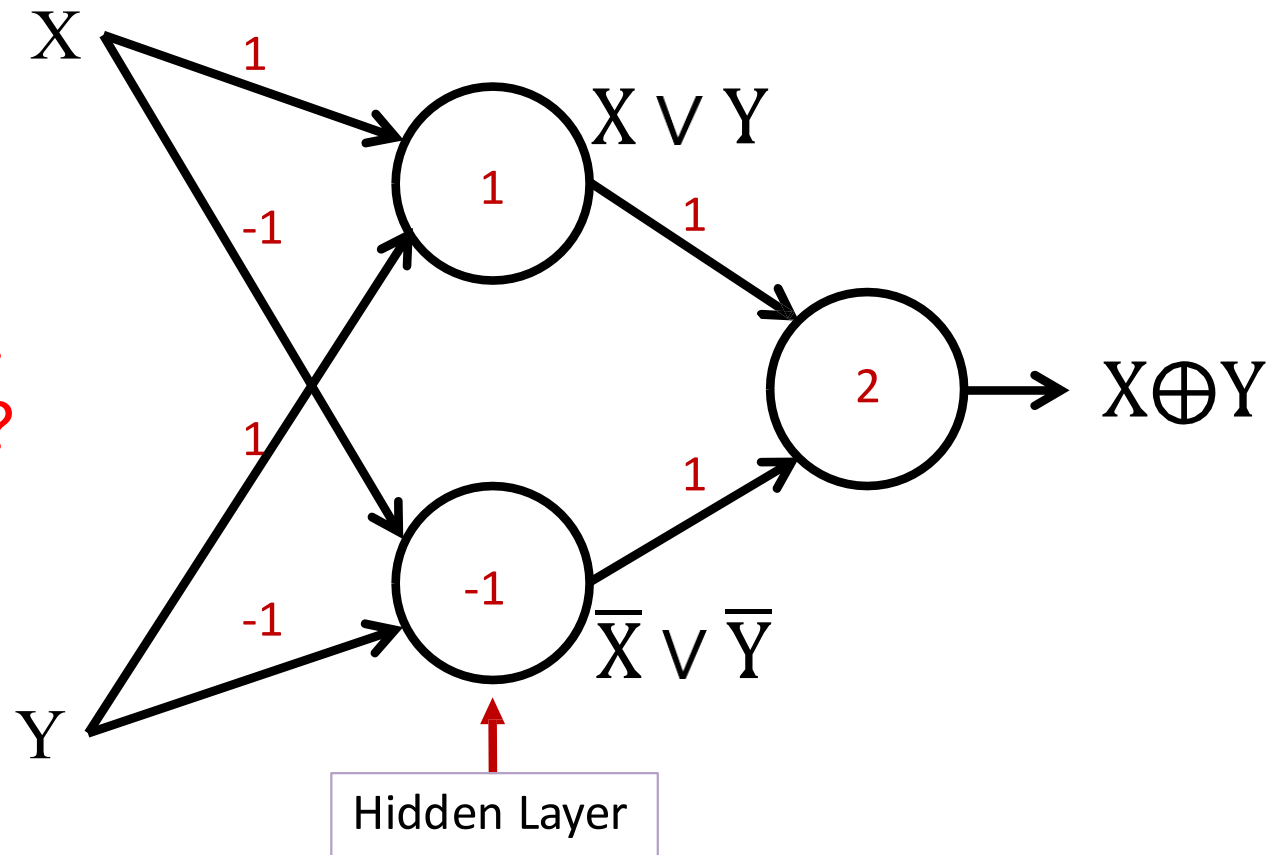
$$w_{t+1} = w_t + (y - \hat{y})x$$

label Prediction

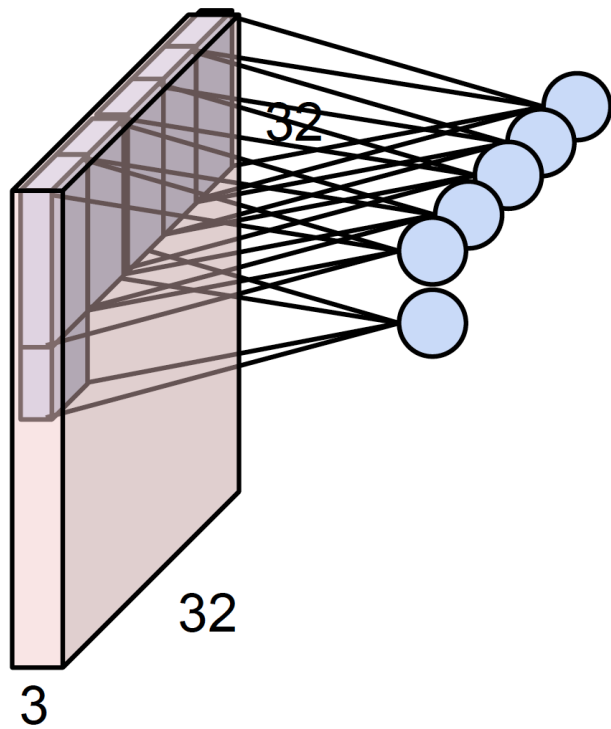


XOR and Multi-layer Perceptron (MLP)

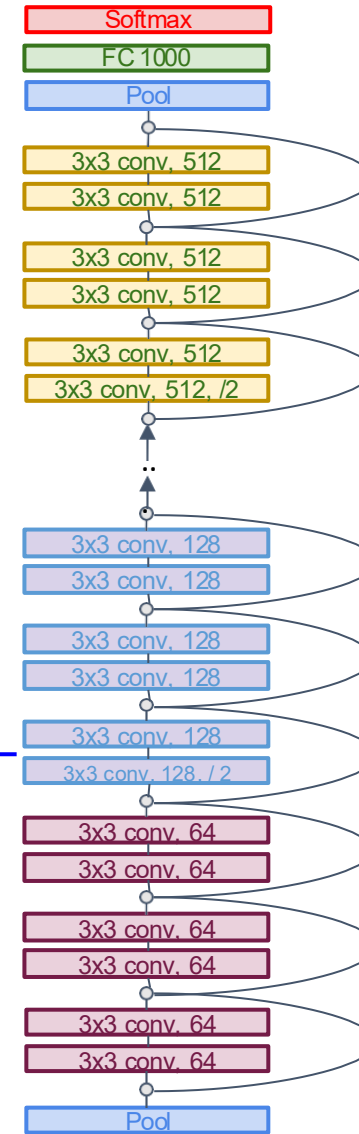
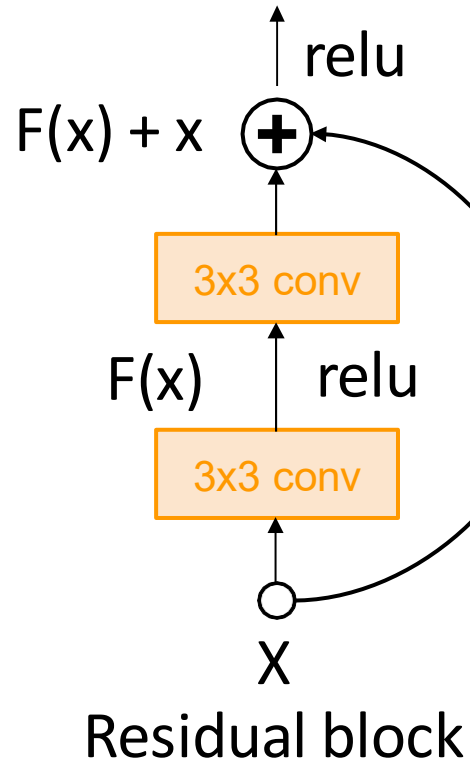
Solution exists...
But how to train?



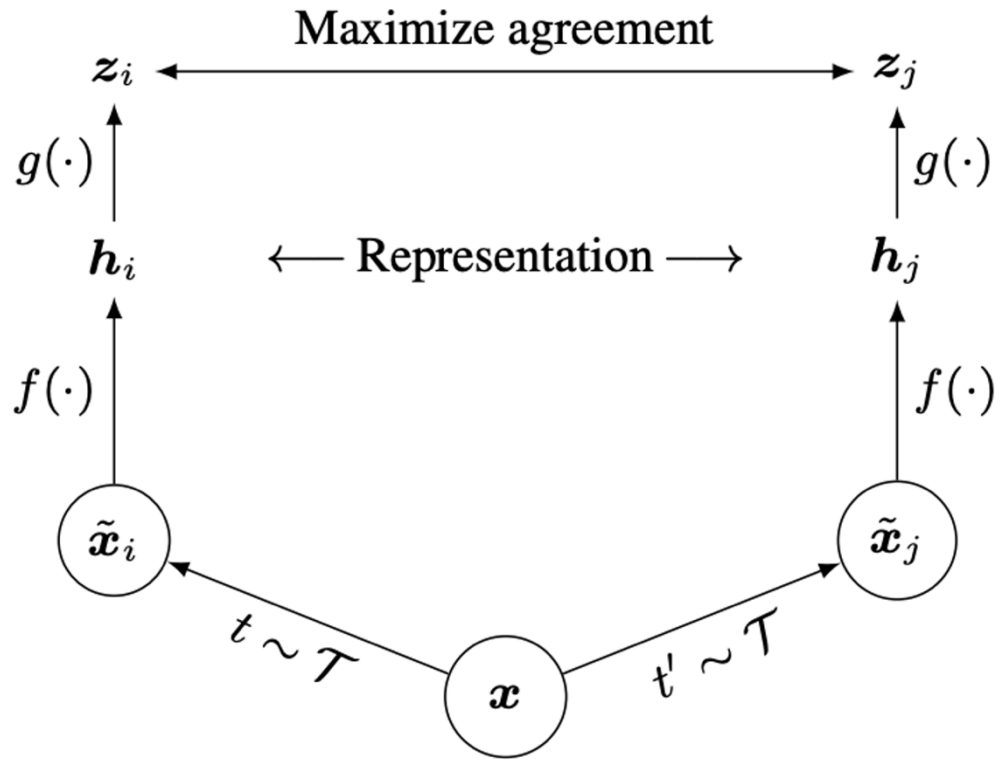
Convolutional Layer



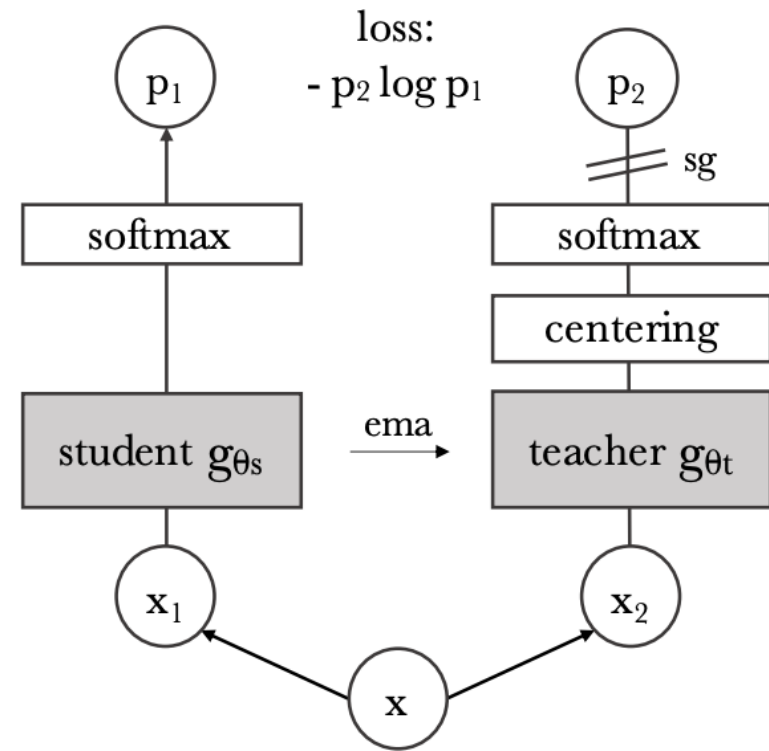
ResNet (Skip Connections)



Self-supervised Models

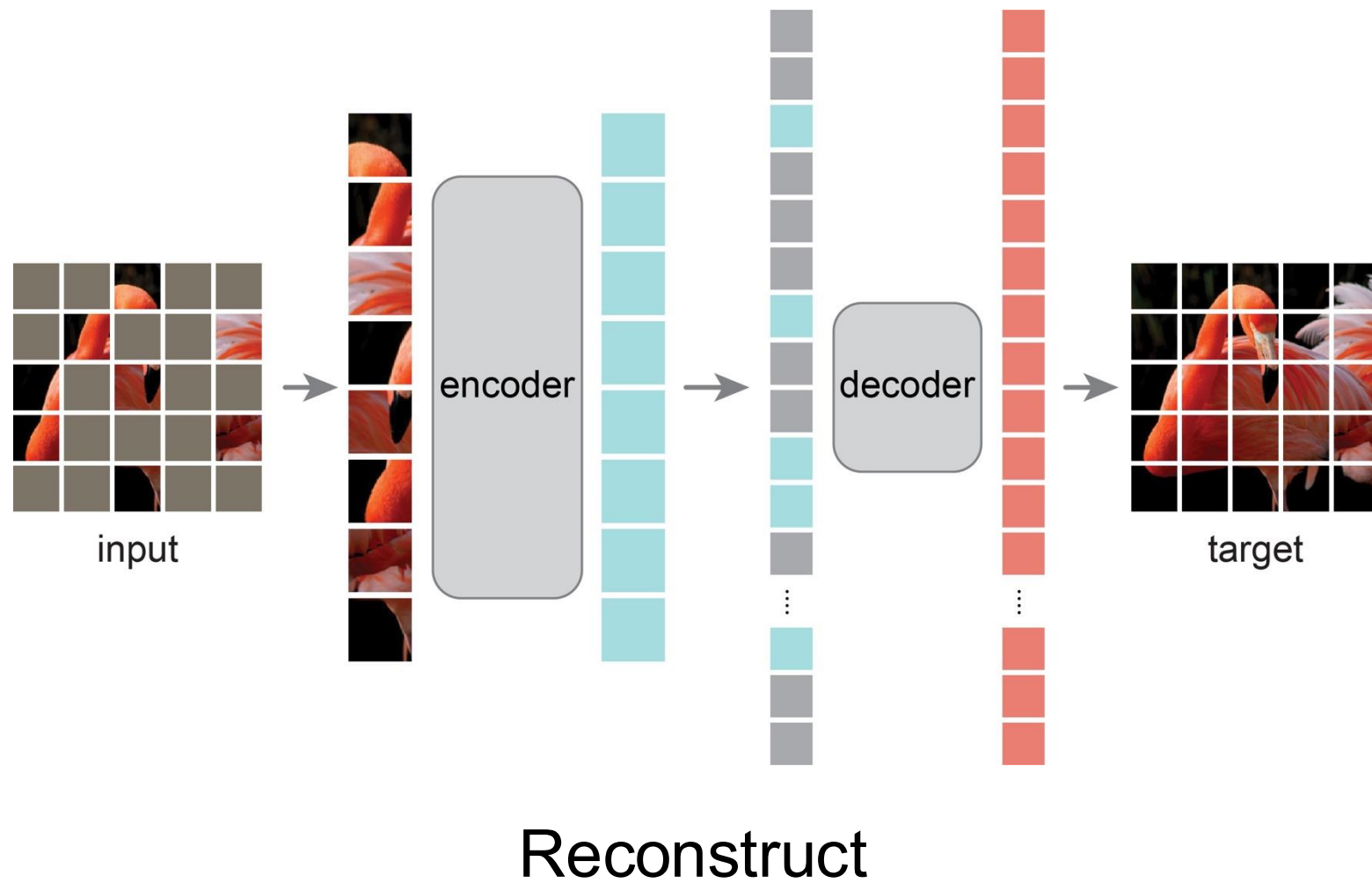


SimCLR



DINO

Reconstruction objective



Sequence-to-Sequence with RNNs

Input: Sequence $x_1, \dots, x_T$

Output: Sequence $y_1, \dots, y_{T'}$

Decoder: $s_t = g_U(y_{t-1}, h_{t-1}, c)$

Encoder: $h_t = f_W(x_t, h_{t-1})$

From final hidden state predict:

Initial decoder state s_0

Context vector c (often $c=h_T$)

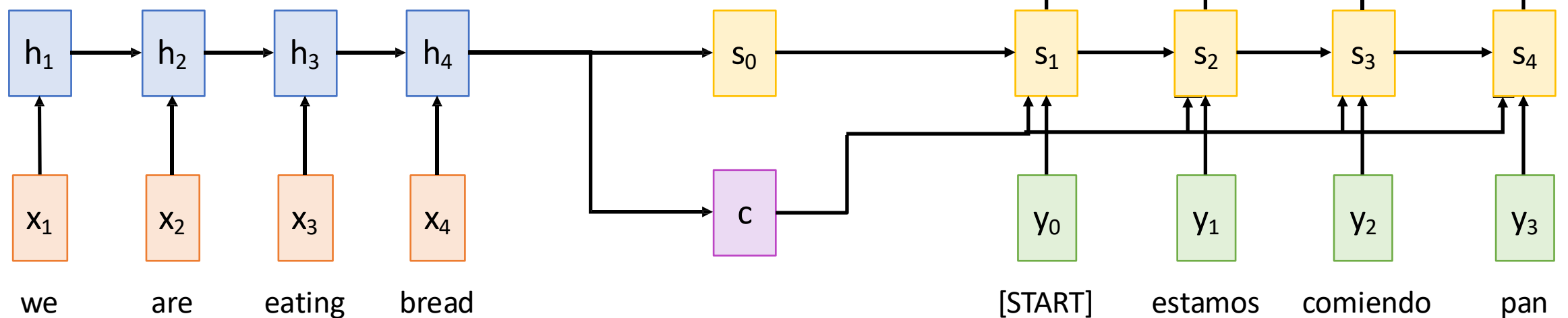


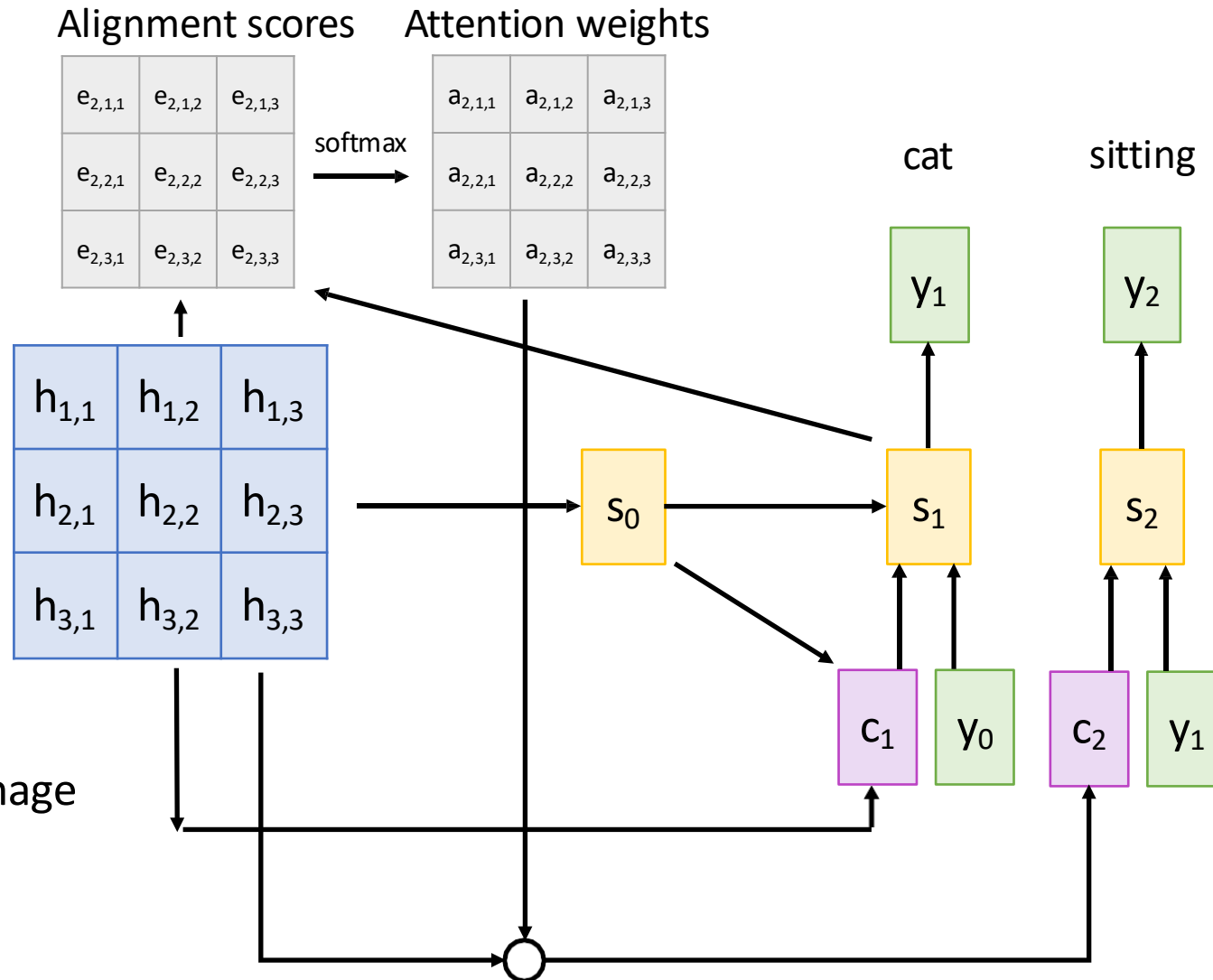
Image Captioning with RNNs and Attention

$$e_{t,i,j} = f_{\text{att}}(s_{t-1}, h_{i,j})$$
$$a_{t,:,:) = \text{softmax}(e_{t,:,:})$$
$$c_t = \sum_{i,j} a_{t,i,j} h_{i,j}$$

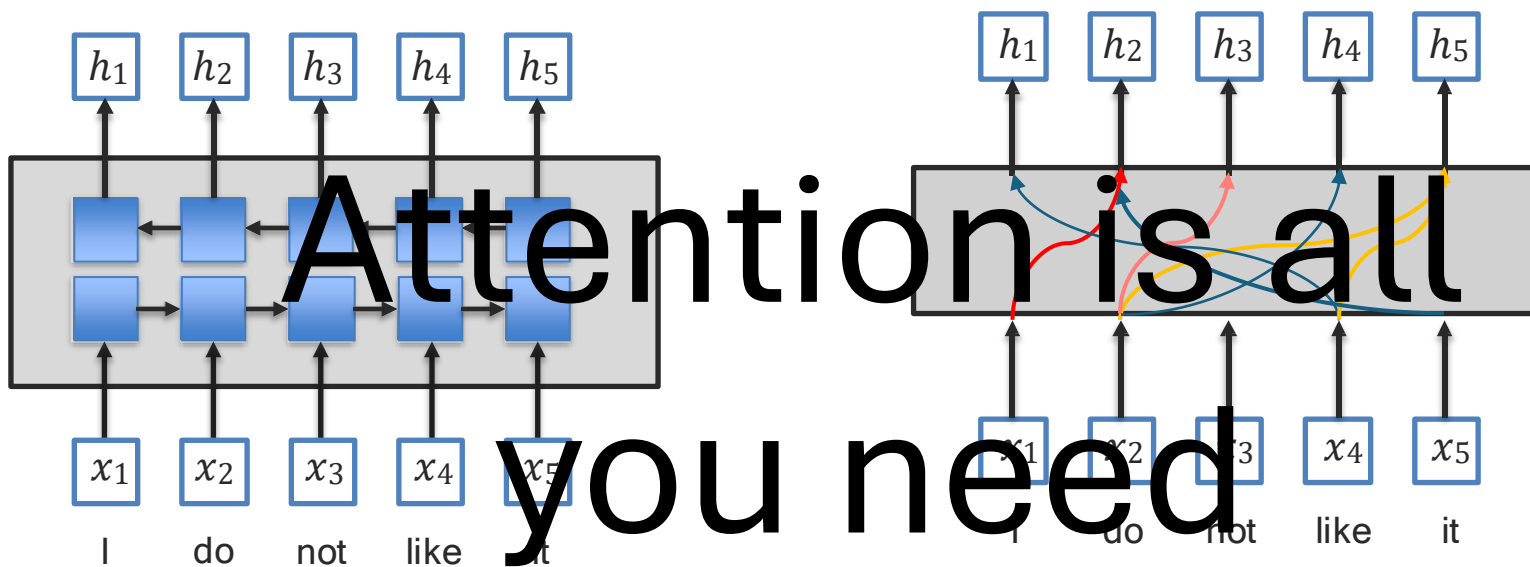


CNN

Use a CNN to compute a grid of features for an image



Attention for sequence representation



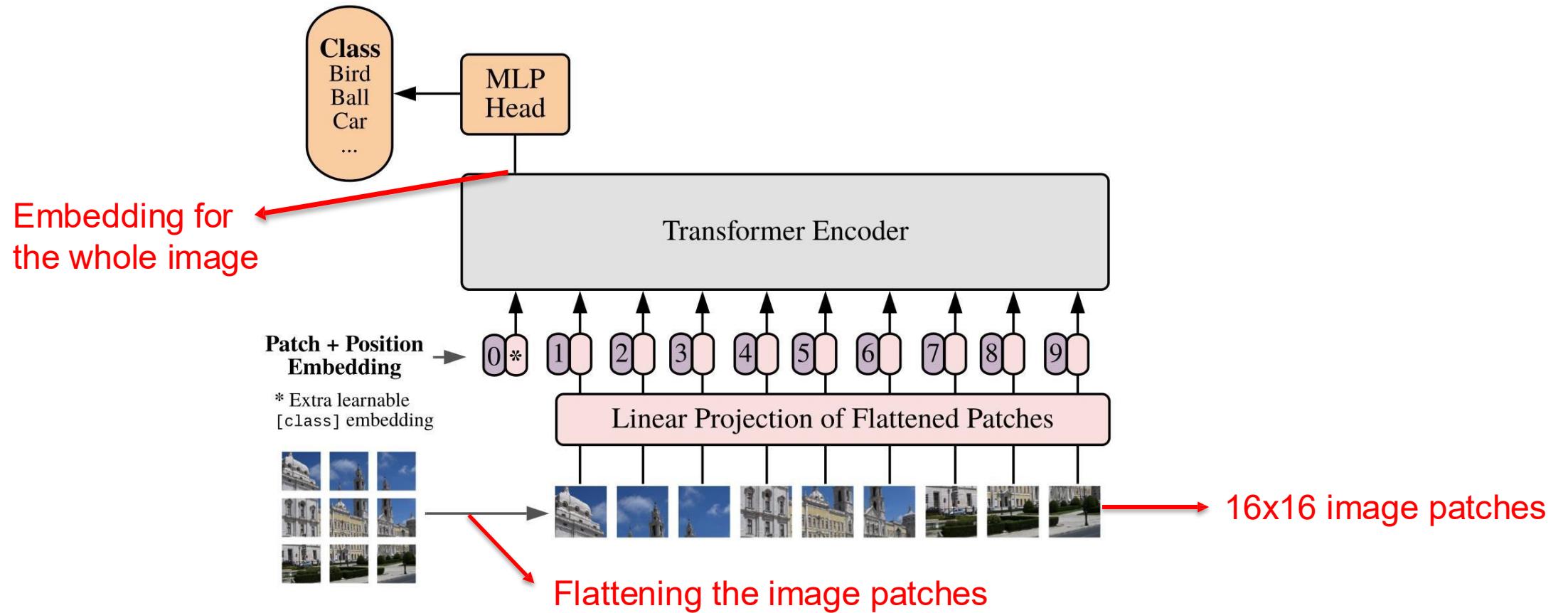
Can process long sequence

But harder to parallelize...

Can be parallelized!

Memory expensive

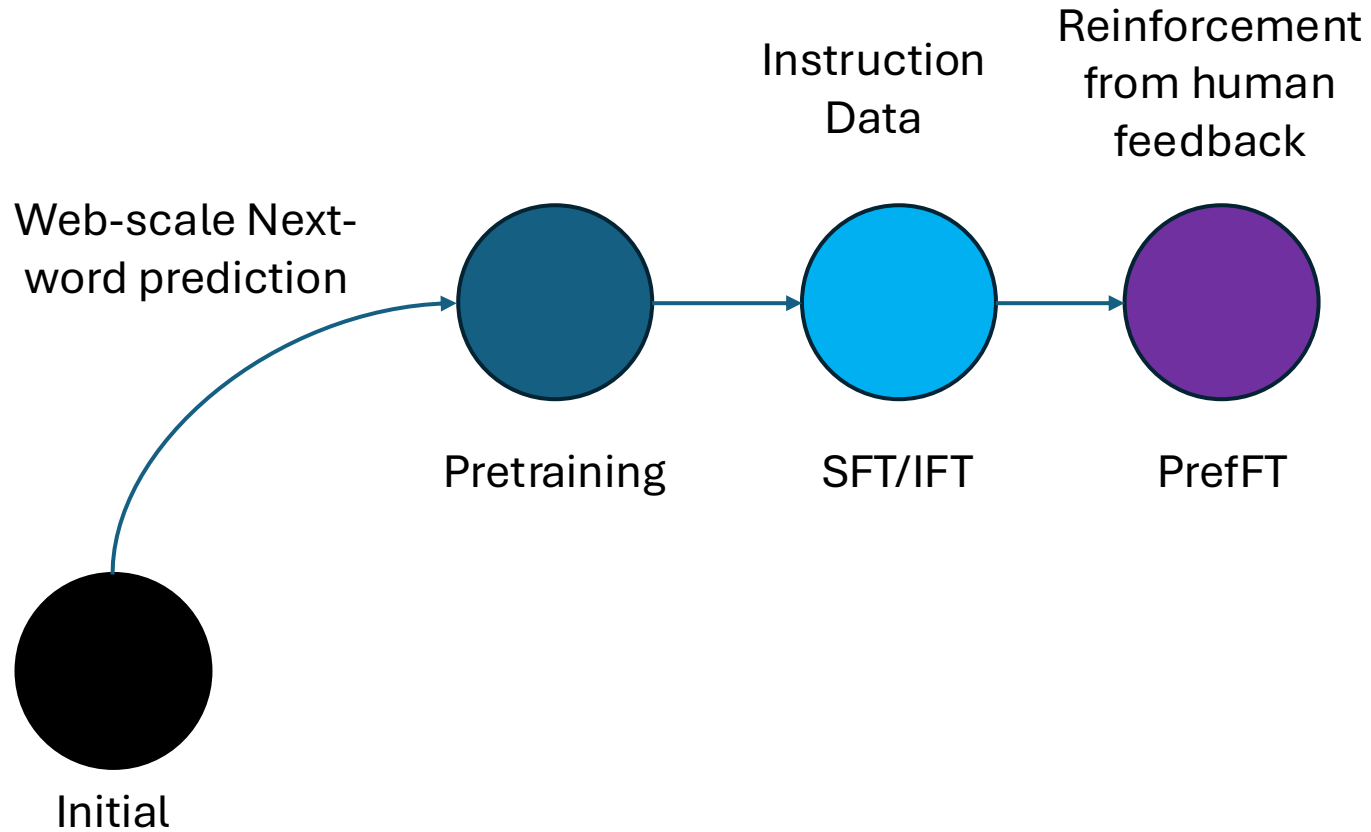
Vision Transformer



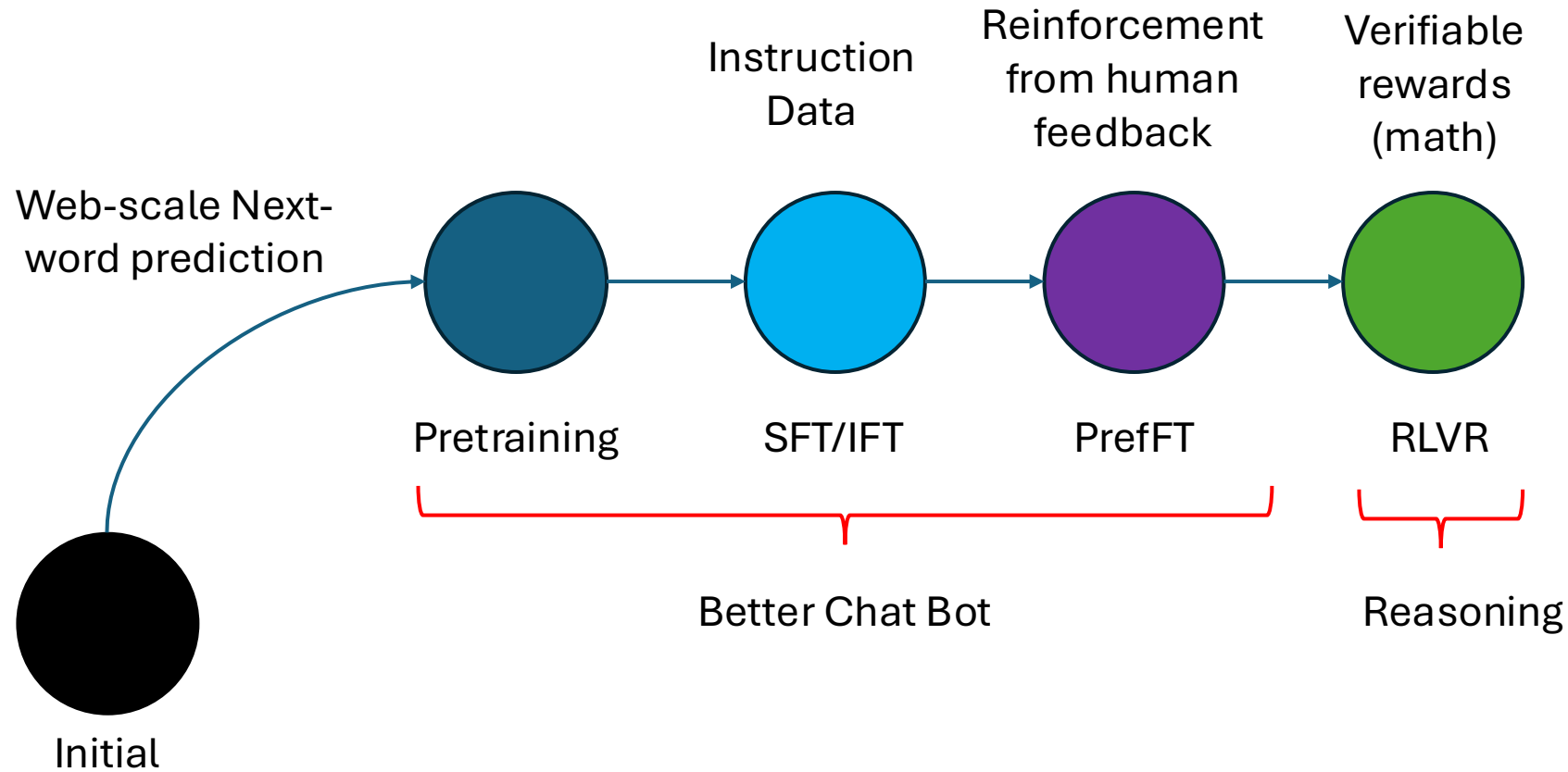
Generating images using diffusion models



Basic steps for Chat GPT like model



Basic steps for O1 like model



What RLVR allows

- Good for reasoning models training
 - reasoning supervision doesn't exist
- **Not mimicking** humans anymore
 - Exploring for a novel solution
- Challenges
 - Rewards are not good enough
 - Process rewards?
 - Very slow to train – bits of information for a long trajectory